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$$1. \quad \bar{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \bar{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \bar{v}_3 = \begin{bmatrix} -1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$$\bar{u}_1 = \bar{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{unit vector } \bar{e}_1 = \frac{\bar{u}_1}{|\bar{u}_1|} = \begin{bmatrix} \sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \\ 0 \end{bmatrix}$$

$$\bar{u}_2 = \bar{v}_2 - \underbrace{\text{proj}_{\bar{u}_1}(\bar{v}_2)}_{\downarrow}$$

$$= \frac{\bar{v}_2 \cdot \bar{u}_1}{|\bar{u}_1|^2} \bar{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\bar{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{unit vector } \bar{e}_2 = \frac{\bar{u}_2}{|\bar{u}_2|} = \begin{bmatrix} 0 \\ \sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \end{bmatrix}$$



$$\bar{u}_3 = \bar{v}_3 - \text{proj}_{\bar{u}_1}(\bar{v}_3) - \text{proj}_{\bar{u}_2}(\bar{v}_3)$$

$$= \begin{bmatrix} -1/2 \\ 1/2 \\ 1/2 \\ -1/2 \end{bmatrix}$$

set of orthonormal vectors is: $\bar{u}_1, \bar{u}_2, \bar{u}_3$

$$\left\{ \begin{bmatrix} \sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ \sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 1/2 \\ 1/2 \\ -1/2 \end{bmatrix} \right\}$$

$$(\bar{u}_1)_{\text{matrix}} \bar{u}_1 = \bar{u}_1$$

$$\begin{bmatrix} \sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \\ 0 \end{bmatrix} = \bar{u}_1$$

$$(\bar{u}_2)_{\text{matrix}} \bar{u}_2 = \bar{u}_2$$

$$\begin{bmatrix} 0 \\ \sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ \sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \end{bmatrix} = \bar{u}_2$$



$$2. a. \bar{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \bar{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \bar{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\bar{u}_1 = \bar{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$\text{Unit vector } \bar{e}_1 = \frac{\bar{u}_1}{|\bar{u}_1|} = \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}$$

$$\bar{u}_2 = \bar{v}_2 - \text{proj}_{\bar{u}_1}(\bar{v}_2)$$

$$= \begin{bmatrix} -4/3 \\ -2/3 \\ 4/3 \end{bmatrix}$$

$$\text{Unit vector } \bar{e}_2 = \frac{\bar{u}_2}{|\bar{u}_2|} = \begin{bmatrix} -2/3 \\ -1/3 \\ 2/3 \end{bmatrix}$$

$$\bar{u}_3 = \bar{v}_3 - \text{proj}_{\bar{u}_1}(\bar{v}_3) - \text{proj}_{\bar{u}_2}(\bar{v}_3)$$

$$= \begin{bmatrix} 2/9 \\ -2/9 \\ 1/9 \end{bmatrix}$$

$$\text{Unit vector } \bar{e}_3 = \frac{\bar{u}_3}{|\bar{u}_3|} = \begin{bmatrix} 2/3 \\ -2/3 \\ 1/3 \end{bmatrix}$$



set of orthonormal vectors are

$$= \left\{ \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}, \begin{bmatrix} -2/3 \\ -1/3 \\ 2/3 \end{bmatrix}, \begin{bmatrix} 2/3 \\ -2/3 \\ 1/3 \end{bmatrix} \right\}$$

b. $Q = \begin{bmatrix} 1/3 & -2/3 & 2/3 \\ 2/3 & -1/3 & -2/3 \\ 2/3 & 2/3 & 1/3 \end{bmatrix}, \quad Q^T = \begin{bmatrix} 1/3 & -2/3 & 2/3 \\ -2/3 & -1/3 & 2/3 \\ 2/3 & -2/3 & 1/3 \end{bmatrix}$

$$Q^{-1} = \left[\begin{array}{ccc|ccc} 1/3 & -2/3 & 2/3 & 1 & 0 & 0 \\ 2/3 & -1/3 & -2/3 & 0 & 1 & 0 \\ 2/3 & 2/3 & 1/3 & 0 & 0 & 1 \end{array} \right]$$

$$R_1 = 3R_1$$

$$= \left[\begin{array}{ccc|ccc} 1 & -2 & 2 & 3 & 0 & 0 \\ 2/3 & -1/3 & -2/3 & 0 & 1 & 0 \\ 2/3 & 2/3 & 1/3 & 0 & 0 & 1 \end{array} \right]$$

$$R_2 = R_2 - \frac{2R_1}{3}$$

$$= \left[\begin{array}{ccc|ccc} 1 & -2 & 2 & 3 & 0 & 0 \\ 0 & 1 & -2 & -2 & 1 & 0 \\ 2/3 & 2/3 & 1/3 & 0 & 0 & 1 \end{array} \right]$$

$$R_3 = R_3 - \frac{2R_1}{3}$$



$$= \left[\begin{array}{ccc|ccc} 1 & -2 & 2 & 3 & 0 & 0 \\ 0 & 1 & -2 & -2 & 1 & 0 \\ 0 & 2 & -1 & -2 & 0 & 1 \end{array} \right]$$

$$R_1 = R_1 + 2R_2$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & -1 & 2 & 0 \\ 0 & 1 & -2 & -2 & 1 & 0 \\ 0 & 2 & -1 & -2 & 0 & 1 \end{array} \right]$$

$$R_3 = R_3 - 2R_2$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & -1 & 2 & 0 \\ 0 & 1 & -2 & -2 & 1 & 0 \\ 0 & 0 & 3 & 2 & -2 & 1 \end{array} \right]$$

$$R_3 = R_3/3$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & -1 & 2 & 0 \\ 0 & 1 & -2 & -2 & 1 & 0 \\ 0 & 0 & 1 & 2/3 & -2/3 & 1/3 \end{array} \right]$$

$$R_1 = R_1 + 2R_3, \quad R_2 = R_2 + 2R_3$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/3 & 2/3 & 2/3 \\ 0 & 1 & 0 & -2/3 & -1/3 & 2/3 \\ 0 & 0 & 1 & 2/3 & -2/3 & 1/3 \end{array} \right]$$

$$Q^{-1} = \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ -2/3 & -1/3 & 2/3 \\ 2/3 & -2/3 & 1/3 \end{bmatrix}$$

$$Q^{-1} = Q^T$$